

Answer **all** questions.

- 1 Yeast is a single-cell eukaryotic organism that can be manipulated and grown in the laboratory much like bacteria. Yeast artificial chromosome (YAC) vectors are designed to act like eukaryotic chromosomes.

pYAC3 contains the following components:

1. an autonomously replicating sequence to function as an origin of replication (*ori*),
2. centromere elements (CEN4) for accurate chromosome segregation to daughter cells during cell division,
3. telomeric sequences (TEL) in order for the ends to be replicated correctly and which also prevent the chromosome from being nibbled away by exonucleases,
4. growth markers (*URA3* and *TRP1*) to select positively for chromosome maintenance, and
5. the *SUP4* gene which encodes for the *SUP4-o* suppressor protein.

The *ADE2* yeast gene encodes an enzyme required in the biosynthetic pathway of adenine. *ADE2* mutants produce a red pigment. When pYAC3 is transformed into a *ADE2* mutant yeast cell, the *SUP4-o* protein suppresses the *ADE2* mutation and white colonies are produced.

Fig. 1.1 shows the pYAC3 vector and the cloning strategy used with pYAC3.

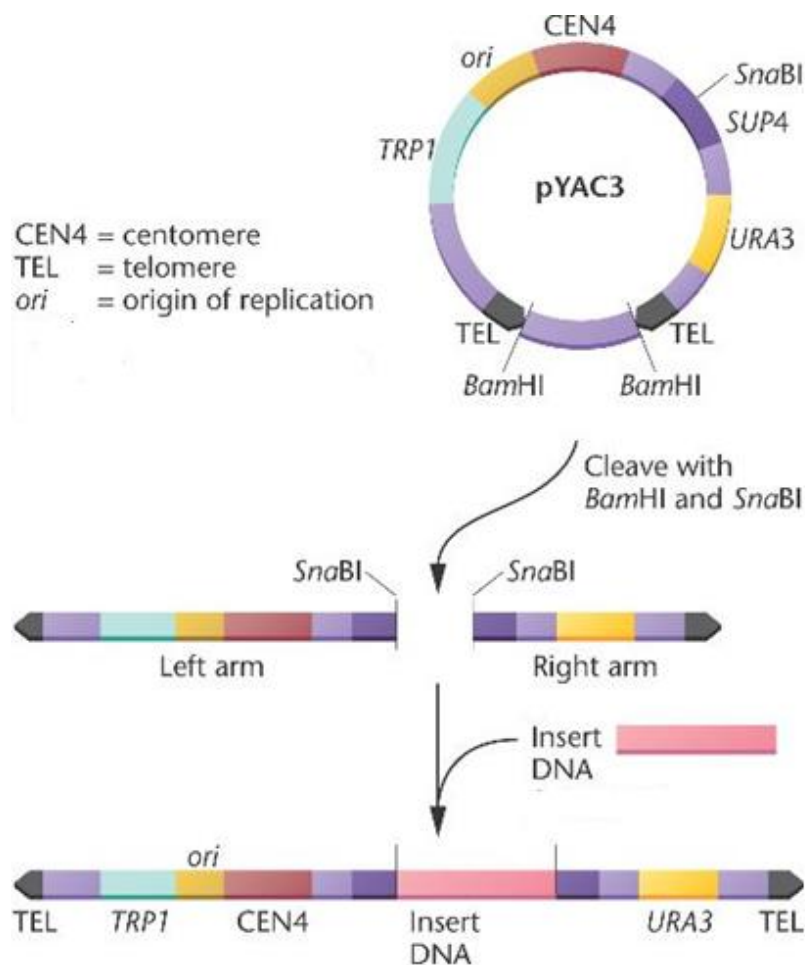


Fig. 1.1

(a) With reference to the preamble and Fig. 1.1,

(i) explain how recombinant *ADE2* mutant yeast cells can be selected.

[3]

Several important mammalian genes such as the human cystic fibrosis gene, which is 250kb in length, have been cloned using the pYAC3 vector.

Using this information and your own knowledge of cloning eukaryotic genes,

(ii) suggest why pYAC3 is a preferred cloning vector for eukaryotic genes compared to bacterial plasmids.

[2]

Fig. 1.2 shows the use of a restriction enzyme EcoRV in a separate cloning experiment.

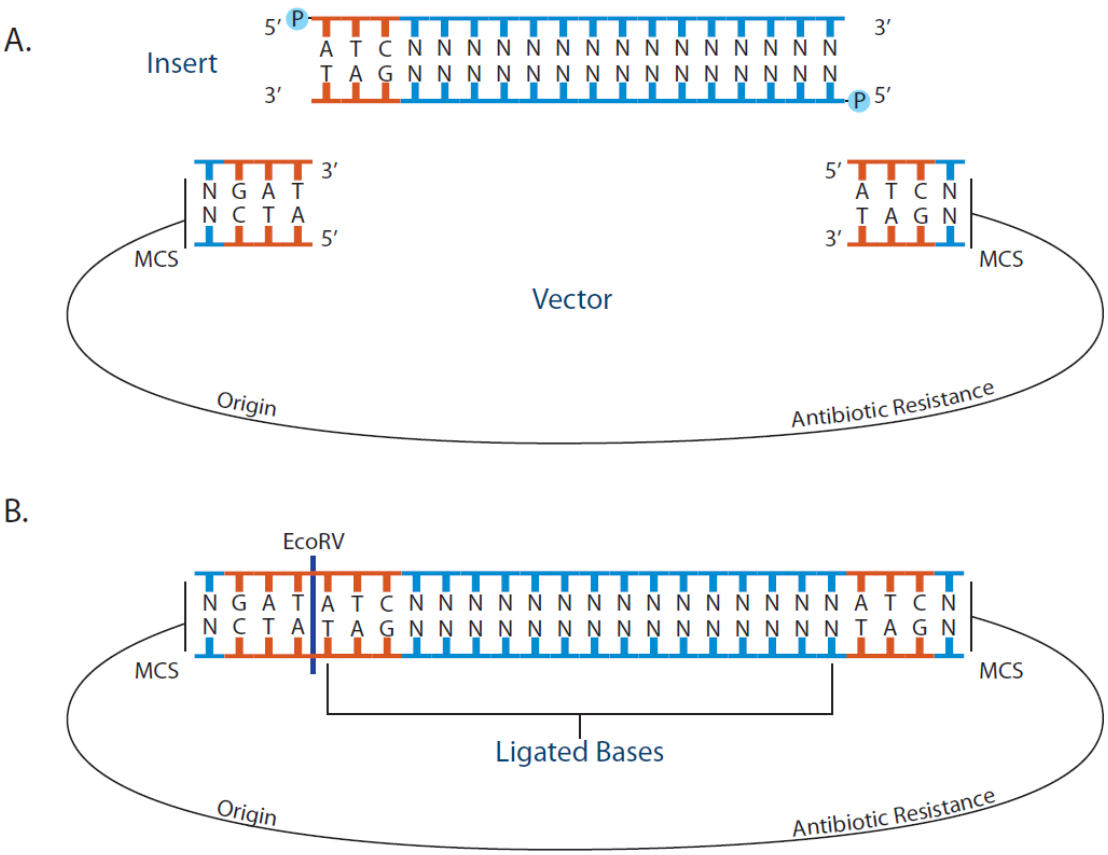


Fig. 1.2

(b) Using the information in Fig. 1.2, explain how a foreign gene can be cloned into the bacterial plasmid shown.

[3]

Transformation is then carried out using competent *E. coli*.

- (c) Outline the additional steps required if the objective of cloning is to harvest the protein products.

[3]

[Total: 11]

- 2 Osteogenesis imperfecta (OI) is a heterogeneous group of genetic disorders that affect connective tissue integrity. The symptoms of OI include fragile bones, osteoporosis, easy bruising and loose joints. The cause of the disease is linked to mutations in the genes that encode polypeptide chains that comprise type I collagen molecules, the major structural protein of bone.

Table 2.1 shows the mode of inheritance and clinical characteristics of type I and type II OI.

Table 2.1

Type	Mode of inheritance	Clinical characteristics
I	Autosomal dominant	Osteoporosis, mild skeletal deformities, mild short stature, easy bruising
II	Autosomal dominant	Lethal form in embryonic development, severe skeletal deformities, beaded ribs and fractures

(a) With reference to Table 2.1,

- (i) explain the cause of the differences in clinical characteristics between type I and type II, in terms of the different types of gene mutation.

[2]

- (ii) explain why conventional gene therapy treatment is not suitable for treating OI.

[2]

The cell therapy approach aims to overcome the problems associated with dominant OI mutations by supplying cells that would give rise to osteoblasts, the bone-forming cells, to replace the cells that synthesize abnormal collagen chains.

Osteoblasts are terminally differentiated, non-proliferating cells with a limited lifespan. Osteoblasts arise from mesenchymal stem cells that reside in the bone marrow whereas the osteoclasts, the cells that are involved in bone resorption, are derived from haematopoietic lineage. The ability for bone to regenerate is attributed to stem cells in bone that undergo proliferation and differentiation.

The rationale for cell therapy is that mutant osteoblasts can be replaced by normal osteoblasts derived from the transplanted progenitors. In addition, since the transplanted cells are stem cells, they will self-renew and thus provide treatment for life.

(b) Suggest the meaning of haematopoietic lineage.

[1]

(c) Explain how mesenchymal stem cells may increase bone growth in children with OI.

[2]

Gene therapy was found to be more effective when used in tandem with anti-sense RNA gene silencing technique. The approach involves the use of double-stranded RNA (dsRNA) sometimes referred to as RNA interference (RNAi). The steps are as follows:

1. Bone marrow cells are taken from diseased individuals.
2. Suppression of diseased RNA expression using RNAi.
3. Transfection of therapeutic gene using a retroviral vector.
4. Injection of treated bone marrow cells back into diseased individuals.

(d) (i) Explain the choice of vector used.

[2]

(ii) State the molecular target of gene silencing.

[1]

(iii) Suggest how dsRNA is used to suppress diseased RNA expression.

[2]

(iv) Suggest why the *ex vivo* approach was adopted for gene therapy treatment of OI.

[2]

[Total: 14]

- 3** In 2012, permission was granted for a field trial in the UK of genetically modified wheat, *Triticum aestivum*. The wheat carries a gene, taken from peppermint plants, which results in the wheat leaves releasing a volatile, non-toxic chemical, (E)- β -farnesene (E β f), into the atmosphere.

E β f is not only produced by various species of plants. It is also secreted by aphids when they are disturbed by a predator.

Two experiments have been performed into the effect of E β f on the behaviour of aphids feeding on leaves in closed containers.

Experiment 1

Either 10 cm³ of air from a syringe that contained plant leaves that secrete E β f
or 10 cm³ of air from a syringe with no such leaves
was added to the containers of feeding aphids.

Experiment 2

Either 20 cm³ of air containing 50ng of E β f
or 20cm³ of air containing no E β f
was added to the containers of feeding aphids.

In both experiments, the number of aphids that stopped feeding and moved away from the food leaves was counted. The results are shown in Table 3.1.

Table 3.1

Air added to containers of feeding aphids	Experiment 1		Experiment 2	
	10 cm ³ air that had been in contact with leaves secreting E β f	10 cm ³ air that had not been in contact with leaves secreting E β f	20 cm ³ air containing 50ng E β f	20 cm ³ air containing no E β f
Number of aphids in container	99	113	132	106
Number of aphids that stopped feeding and moved away from the food leaves	54	1	111	0

- (a) (i)** With reference to Table 3.1, discuss the extent to which the results of these experiments support the idea that E β f is an alarm signal for aphids.

[3]

- (ii)** Other experiments show that E β f attracts predators of aphids, such as ladybirds.

Explain how growing genetically modified wheat secreting E β f could increase the yield of wheat.

[3]

- (iii)** Suggest why growing this genetically modified wheat might be acceptable to people who object to the growth of genetically modified insect-resistant maize or cotton.

[3]

In a separate experiment, *Agrobacterium*-mediated transformation was performed on *T. triticum* to produce seven transgenic lines, with enhanced E β f secretion.

Research into two of the cell lines, E β f-1 and E β f-2, showed that there is a distinct genetic difference between them, and both are tagged with a different genetic marker; E β f-1 with genetic marker 1 and E β f-2 with genetic marker 2, as shown in Fig. 3.1.

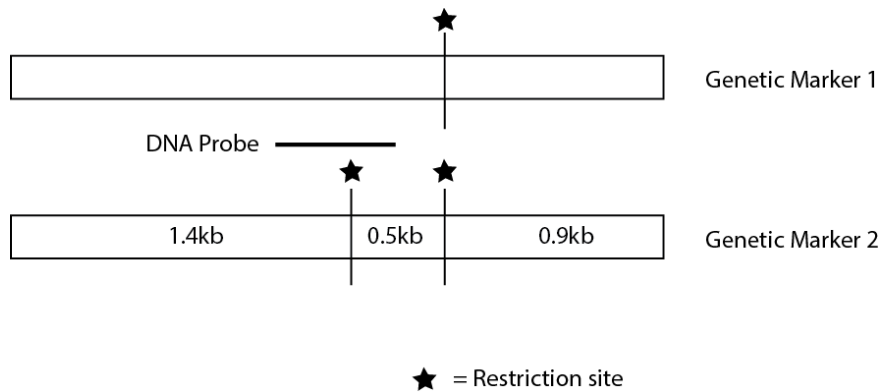


Fig. 3.1

- (b) (i)** Explain how restriction fragment length polymorphism comes about.

[2]

- (ii)** With reference to Fig. 3.1, describe how RFLP can be used to identify cell line carrying E β f-2 gene.

[4]

[Total: 15]

4 Planning Question

Enzymatic reactions are influenced by various factors, such as the presence of inhibitors. There are two main types of inhibitors, the competitive inhibitors and non-competitive inhibitors. The mode of action of the two types of inhibitors differs and both have a different effect on enzyme-catalysed reactions.

It was known that the action of amylase is inhibited by the presence of ions such as copper (Cu^{2+}). Plan an investigation to determine if Cu^{2+} functions as a competitive inhibitor or a non-competitive inhibitor of amylase.

Your planning must be based on the assumption that you have been provided with the following apparatus and materials, as well as other appropriate apparatus and equipment found in the school laboratory.

- 1 mol dm^{-3} amylase enzyme
- 5 g dm^{-3} starch solution
- Benedict's reagent
- 5 mol dm^{-3} Copper Sulphate (CuSO_4) stock solution
- 10 mol dm^{-3} maltose solution
- Glass rods
- White tiles
- Bunsen burner, tripod and wire gauze
- Stopwatch
- Access to water bath
- Other measuring instruments (measuring cylinder, syringes, etc.) and apparatus (beakers, test-tube, etc.)

Your plan should have a clear and helpful structure to include:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it, including scientific reasoning behind procedure.
- Be illustrated by relevant diagram(s) to show, for example, the arrangement of apparatus used.
- Include the layout of result tables and graphs with clear headings and labels.
- Include full details and explanations of the procedures that you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible.
- Include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

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[illegible]

Free response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 **(a)** Describe cystic fibrosis and how it can be treated using a non-viral gene delivery system. [8]
- (b)** Describe how the micropropagation technique is used in cloning and production of herbicide resistant plants. [7]
- (c)** Discuss the social and ethical implications of genetically modified plants using 2 named examples. [5]

[Total: 20]

END OF PAPER